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Original Report

Orbital radiation therapy for Graves' ophthalmopathy: Measuring clinical efficacy and impact

Ezra Hahn MD [a], Normand Laperriere MD, FRCPC [a], Barbara-Ann Millar MBChB, FRCPC [a], James Oestreicher MD [b], Hugh McGowan MD [b], Hatem Krema MD [b], Harmeet Gill MD [b], Dan DeAngelis MD [b], Jeff Hurwitz MD [b], Nancy Tucker MD [b], Rand Simpson MD [b], Caroline Chung MD, FRCPC, CIP [a,*]

[a]*Department of Radiation Oncology, University of Toronto, Toronto, Ontario, Canada*
[b]*Department of Ophthalmology and Vision Sciences, University of Toronto, Toronto, Ontario, Canada*

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Abstract

Purpose: Graves' ophthalmopathy (GO) is an autoimmune condition primarily managed with prolonged courses of glucocorticoids, which can be associated with significant side effects. Orbital radiation therapy (RT) is an alternative treatment that has shown variable efficacy in improving orbital and visual symptoms. In this study, the therapeutic benefit of RT was evaluated in terms of patient's ability to taper their corticosteroid requirements, which may better reflect the proposed mechanism of RT and provide a clinically relevant response endpoint.

Methods and materials: This is a retrospective review of consecutive patients treated with orbital RT for GO between 2000 and 2010 at a single tertiary hospital with a dedicated ocular radiation therapy clinic. The primary measure of treatment response was defined as the ability to taper glucocorticoids following RT without any further exacerbation of orbitopathy symptoms. Additional endpoints including ocular symptoms (diplopia, proptosis, visual acuity, extraocular movement) and need for surgical intervention were reported.

Results: Of 86 eligible patients, with a mean follow-up of 9.3 months, 81 (94%) patients responded to RT. Of patients taking corticosteroids at baseline, 91% were able to taper off corticosteroids completely and the remaining patients had decreased their doses by 83%. Diplopia, visual acuity, and extraocular movements improved in 29%, 81%, and 58% of patients, respectively. The median reduction in proptosis was 2.5 mm and 2 mm in the left and right eyes, respectively (range, −18 mm to 23 mm).

Conflicts of interest: None.

* Corresponding author. 610 University Ave, 5-974 Toronto, Ontario, Canada, M5G 2M9.
E-mail address: Caroline.chung@rmp.uhn.on.ca (C. Chung).

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Conclusions: Orbital RT is a generally well-tolerated treatment that helps minimize the dose and duration of corticosteroid therapy for patients with GO while improving ocular symptoms, including proptosis and diplopia. Prospective research should consider using corticosteroid requirement as a measure of response to orbital RT for GO.
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Introduction

Graves' ophthalmopathy (GO) is clinically present in 20%-25% of patients at the time of their diagnosis of hyperthyroidism.[1] A significantly larger proportion of patients with hyperthyroidism, up to 71%, have radiologic evidence of extraocular muscle enlargement on magnetic resonance imaging consistent with GO but no bothersome reported symptoms.[2] In the general population, the estimated incidence of GO is 16 women and 3 men per 100,000 population per year.[3]

The natural history of GO is widely variable with spontaneous remission seen in some patients and persistent or progressive disease seen despite medical treatment in others.[4] Glucocorticoid therapy has been the primary treatment for patients suffering from symptomatic GO.[5,6] Studies comparing oral and intravenous corticosteroids have reported varying adverse event rates among groups with weight gain, hypertension, and development of cushingoid features as common adverse events in the oral corticosteroid group and flushing, dyspepsia, and palpitations reported in the intravenous group.[7-10] The risk of any adverse event was related to the dose and duration of glucocorticoid treatment.[11]

Given the adverse events related to glucocorticoid treatment, orbital radiation therapy has been investigated as an alternative therapy for GO. However, studies to date have reported a wide range of responses to radiation therapy for GO.[12-15] This is, at least in part, due to variable endpoint measures of response to radiation therapy and inclusion of patients at all stages of disease in these studies. Response endpoints used in prior studies include reduction in proptosis or orbital muscle bulk, and improvement in diplopia. Although these endpoints are clinically important, as they do not reflect the proposed mechanism of radiation therapy (RT), a suppression of the inflammatory process, use of these measures as primary endpoints may underestimate the efficacy of radiation therapy. This study defines response to RT as the ability of patients to taper glucocorticoids following RT without further exacerbation of their orbitopathy symptoms. This endpoint may better reflect the proposed effect of RT to suppress the inflammatory process and provides a clinically relevant response measure that accounts for the association between the dose and duration of glucocorticoid therapy and its related toxicities.

Methods and materials

This is a retrospective review of consecutive patients treated with orbital RT for GO between 2000 and 2010 at a single tertiary hospital with a dedicated ocular radiation therapy clinic. Research was completed in accordance with local research ethics board requirements. The following baseline data were collected: dose of steroids at consultation and initiation of RT; prior radioactive iodine treatment; and orbital decompression surgery, diplopia, proptosis, visual acuity, and extraocular movements. Follow-up data included steroid dose based on patient history and prescription records, diplopia reported by the patient history and observed during the physical exam, proptosis based on clinical exam including exophthalmometer readings, visual acuity, and extraocular movements based on physical exam, and any ocular surgery (decompression, muscle-corrective, and oculoplastic surgery). Standard clinical follow-up generally included assessment after RT at 3-month intervals for the first 18 months then yearly. However, based on patient preference some patients were followed initially in the Ocular Radiotherapy Clinic then discharged to their ophthalmologist or endocrinologist with a corticosteroid tapering plan and instructions to the patient and referring physician to return to the clinic upon recurrence of symptoms at any point after RT. Patients with either follow-up regimen were considered to have full follow-up. Patients who did not return for their follow-up appointments and did not have a clear corticosteroid tapering plan while being followed by their referring physician, were censored due to missing information in the statistical analysis.

The primary endpoint was response to RT, defined as the ability to wean down or off glucocorticoids at 6 months following completion of RT without any further GO flares after 6 months from completion of RT. "Flare" was defined as a need for further intervention (increase in steroids or surgery) due to increased symptoms any time beyond 6 months after completion of RT. Descriptive statistics were used to report steroid requirements and any further incidence of flares, as well as endpoints that have been reported previously in publications including changes in diplopia, proptosis, visual acuity, and extraocular movements.

Table 1 Characteristics of the study population at initiation of radiation therapy

Characteristic	No. (%)
Sex	
Male	38 (44)
Female	48 (56)
Prior radioactive iodine treatment	26 (30)
Steroid use	
Any steroids	69 (80)
On steroids at start of RT	60 (70)
Average daily prednisone dose at start of RT (range)	38 mg (5 mg-100 mg)
Symptoms	
Diplopia	52 (84)
Proptosis	59 (95)
Decreased visual acuity	51 (88)
Restricted extraocular movements	64 (82)
Orbital decompression prior to RT	19 (22)

RT, radiation therapy.

Results

There were 86 consecutive patients who met eligibility criteria, with 38 men (44%) and 48 women (56%) with median ages of 50 (range, 34-81) and 54 (range, 35-86) years, respectively (Table 1). All 86 patients received a total dose of 2000 cGy delivered in 10 daily fractions over 2 weeks to both orbits (except 1 patient who only received radiation to her right orbit [Fig 1]). The majority of patients (n = 69) had been taking prednisone for varying durations prior to the initial radiation therapy consultation and were generally referred due to difficulty tapering their corticosteroids due to exacerbation of orbitopathy symptoms upon tapering. The mean dose of prednisone at the time of initial consultation was 47.5 mg daily (range, 7.5 mg-100 mg). Among the 13 (15%) patients who had never used steroids prior to consultation, a variety of contraindications (including glaucoma, hypertension, and diabetes) to steroids were noted.

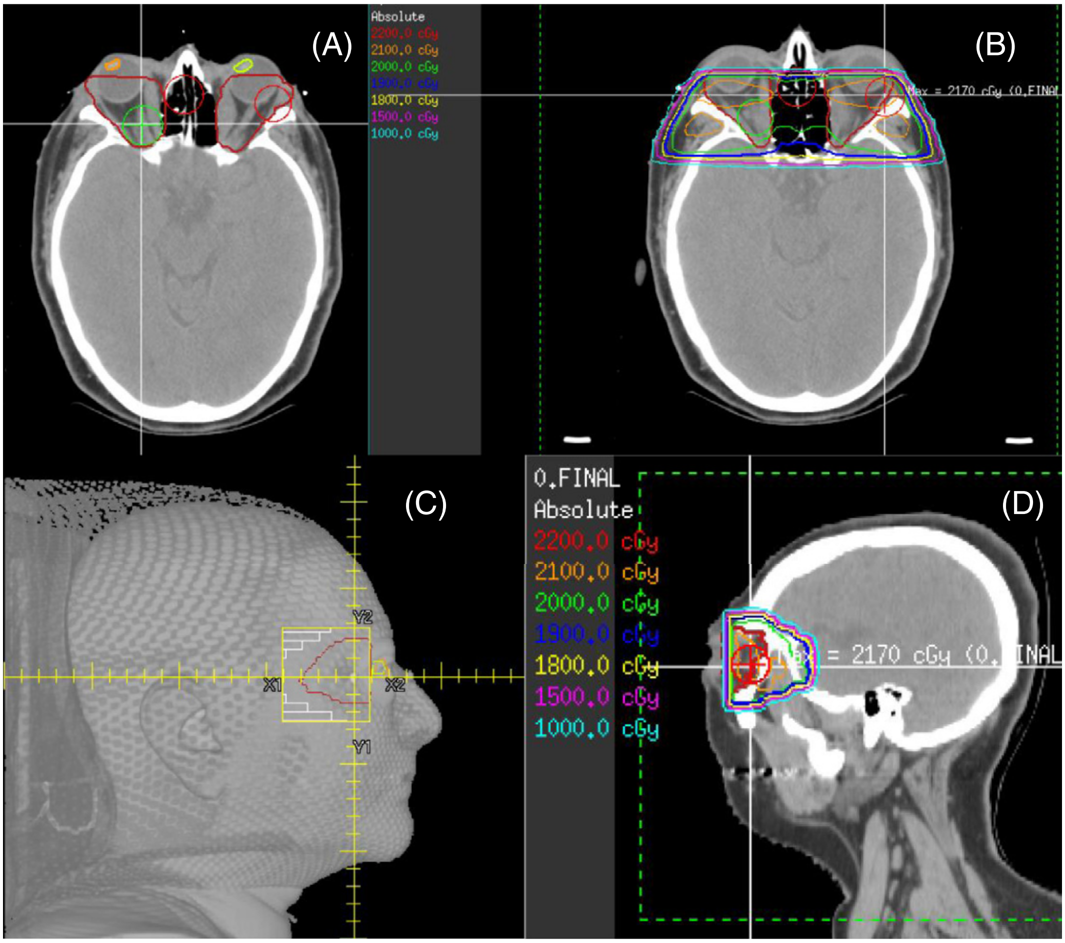

Figure 1 Representative images of a radiation therapy plan for Graves' ophthalmopathy. (A) Axial computed tomographic (CT) image showing the clinical target volume encompassing the extraocular muscles delineated in red. (B) Axial CT image showing the radiation dose distribution through both orbits, sparing the lenses and anterior chambers of both orbits. (C) Sagittal image of the skin rendering of the patient in an immobilization mask with field borders. (D) Sagittal CT image showing the radiation dose distribution focused on the orbital muscles, sparing the anterior globes and brain.

Baseline assessment prior to radiation therapy

At the start of radiation therapy, 60 patients were taking prednisone at a mean dose of 38 mg (range, 5 mg-100 mg) per day. At baseline, 51 patients had impaired visual acuity, 62 patients had documentation of diplopia of which 52 (84%) complained of this symptom, and 62 patients had documented proptosis of which 59 (95%) had exophthalmometer measurements. Documented limitation in extraocular movements was reported in 78 patients of which 64 (82%) reported symptomatic limitation in their extraocular movements. Nineteen patients had orbital decompressive surgery prior to receiving orbital radiation therapy, mainly due to vision loss despite medical intervention.

Radiation treatment response and toxicities

With an average in-clinic follow-up of 9.3 months (range, 1-36.5 months), 81 (94%) patients responded to RT. Five patients (6%) failed RT and required further intervention for progressive symptoms following RT. There were no patients with a documented late flare of GO after 18 months following orbital RT (Table 2). There were no grade 2 or greater acute or late radiation-related toxicities observed within this study cohort.

Among patients with full follow-up who took corticosteroids at induction of RT, 90.9% completely weaned off corticosteroids by the end of follow-up. The average percent decrease of their steroid dose, relative to their dose at induction of radiation therapy, was 96.1% (Fig 2). Of the patients who were unable to fully taper steroids, their steroid requirements were reduced to 26.7% of their doses at the start of RT. Only 1 of these patients required a small increase in steroid dose at the end of follow-up. There was incomplete documentation of corticosteroid doses in 11 patients and they were censored in the statistical analysis; 1 of these patients was transitioned to palliative care due to development of rapidly progressive anaplastic thyroid cancer.

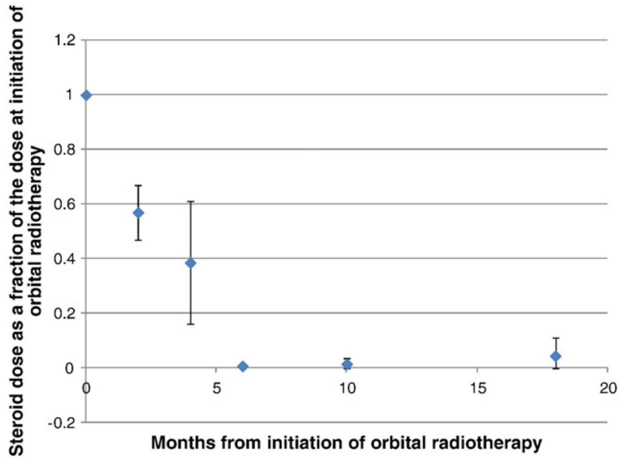

Figure 2 Proportional corticosteroid requirement following radiation therapy. The data points represent the mean steroid dose of patients at a given month as a proportion of their total steroid dose at initiation of orbital radiation therapy. Error bars represent 95% confidence intervals. The increase in range of the confidence interval at 18 months is due to 1 patient whose steroid dose increased by the end of follow-up.

In the subgroup of 19 patients who received orbital decompressive surgery prior to orbital RT, 5 patients completely tapered off corticosteroids and 2 patients had a 50% reduction in their corticosteroid doses by last follow-up. Seven patients had no clear documentation of corticosteroid use and 5 patients were not taking corticosteroids at initiation of RT.

Table 2 Steroid therapy after orbital radiation therapy

Characteristic	No. (%)
Able to discontinue steroids	40 (91)
Unable to discontinue steroids	4 (9)
Average percent decrease in steroid dose	96.10%
Average percent decrease in steroid dose in patients who were unable to fully reduce their steroid dose	83.30%
Off all steroids by 6 mo	38 (84.4)
Failed radiation therapy	5 (7.5)
Flare of GO at >18 mo	0
Patients censored from statistical analysis due to missing information	11 (18)

Table 3 Symptom outcomes postorbital radiation therapy (RT)

Outcome	No. (%)
Diplopia	
Improved	16 (31)
Worsened	2 (4)
Proptosis	
Median improvement in the left eye (range)	2.5 mm (−18 mm to 23 mm)
Median improvement in the right eye (range)	2.0 mm (−18 mm to 23 mm)
Visual acuity	
Improved	30 (81)
Worsened	6 (16)
Extraocular movement restriction	
Improved	29 (58)
Worsened	5 (10)
Orbital decompression surgery after RT	5 (6)
Eyelid oculoplastic surgery after RT	14 (16)
Muscle corrective surgery after RT	16 (19)

Ocular symptoms

Of the 52 patients who reported diplopia at baseline, 15 patients (29%) noted an improvement after RT (Table 3). Two patients reported the development of new diplopia following RT. The median reduction in proptosis during follow-up, measured using an exophthalmometer, was 2.5 mm and 2 mm in the left and right eyes, respectively. Of the 37 patients with decreased visual acuity at baseline, 30 patients (81%) had improvement in visual acuity following RT. Twenty-nine of 50 (58%) patients with documented restriction in extraocular movements had improvement following RT but 5 patients reported increased limitation in extraocular movement in at least 1 additional direction following RT.

Surgical intervention

A total of 29 patients had surgical intervention after RT. The majority of patients received muscle-corrective (14 patients). There were 5 patients who received orbital decompression, 2 of whom required this procedure urgently following RT failure.

Discussion

Our study evaluated the efficacy of RT in terms of the patient's ability to taper glucocorticoids without any further exacerbations of orbitopathy symptoms as the primary outcome measure. It has been recognized that prolonged use of high-dose corticosteroid therapy can significantly impact quality of life and treatment-related morbidity. Long-term and high-dose glucocorticoid therapy can impact nearly every system, resulting in a number of common bothersome symptoms and medical conditions including insomnia, acne, skin thinning, weight gain, and changes in appearance (cushingoid features, hirsutism), mood problems, proximal muscle weakness, hyperglycemia, and hypertension.[16-18] Over 90% of patients taking glucocorticoids report experiencing at least 1 adverse event related to therapy and 55% of patients reported at least 1 adverse event that was very bothersome.[19] In our study, the majority of patients on corticosteroids prior to RT had dramatic reductions in corticosteroid requirements, including complete discontinuation of corticosteroids in 91% of patients following RT. This was in keeping with studies that analyzed steroid requirements as secondary outcome measures of RT efficacy at specified time points.[20,21] This highlights the role of RT to stabilize the disease process and minimize the duration of corticosteroid therapy and its associated side effects.

The role of radiation therapy in the treatment of Graves' ophthalmopathy has remained controversial due to incongruent trial results. Two contributing factors to these inconsistent trial results include the variable primary outcome measures used to evaluate the efficacy of RT and inclusion of patients at all stages of their disease. In part, this stems from our limited understanding of the underlying pathogenesis and natural history of Graves' ophthalmopathy and the therapeutic mechanisms of RT in this condition.

Graves' ophthalmopathy is an autoimmune disease with the following 3 main phases: active phase; plateau or static phase; and inactivation phase.[22] The active phase is characterized by periorbital and orbital inflammation, leading to volume increase of extraorbital muscles and orbital adipose tissue. This pathophysiology accounts for the clinical presentation of proptosis, diplopia, extraocular movement restriction, and compressive optic neuropathy with a risk of decreased visual acuity. It is during this active phase that medical treatment, corticosteroids and RT, is thought to be most effective, highlighting the importance of early intervention, possibly within the first year of onset of symptoms.[22] Remission during the inactivation phase can occur spontaneously but the patient can often be left with residual cosmetic or functional changes. Therefore, patients in the inactivation phase will often require surgery to have improvements in their orbital and visual symptoms and quality of life.[22] This emphasizes the importance of early consideration of RT in patients with GO who are not responding to corticosteroids and consideration of outcome measures other than improvements in orbital and visual symptoms to evaluate the response of RT.

A recent review on the efficacy and safety of orbital RT for GO concluded that the combination of orbital RT and oral glucocorticoids is superior to either alone; also, that most interventions, including orbital RT, are most effective during the active phase of disease such that the inclusion of patients with late GO in intervention clinical trials can underestimate the efficacy of intervention.[23] Consistent with our study findings, this review concluded that RT significantly reduced the need for further treatment, including further steroid or immunosuppressive therapy.[23,24]

Prospective studies comparing RT with sham irradiation and comparing RT with corticosteroids for GO reporting variable responses to RT have generally included patients at all phases of disease and have largely aimed at measuring response to ocular symptoms.[13,14,24] A recent meta-analysis of prospective studies did not show a difference between RT and sham RT, based on changes in the clinical activity score (CAS, a validated scoring system based on signs of inflammation such as pain, redness, swelling, and impaired function), but did show that RT yielded a higher response rate for diplopia with an odds ratio of 4.88.[15] This meta-analysis concluded that orbital RT as a sole treatment modality for GO remains controversial, but RT in combination with corticosteroids has greater efficacy than either treatment alone.[12,15] A recent Cochrane Review of orbital RT for GO,[25] which included only 2 studies in their meta-analysis due to the large heterogeneity of outcome measures in prior trials, concluded that RT was superior to sham RT but RT was not significantly better than oral corticosteroids.[25] The individual studies included in this review suggested that RT in combination with corticosteroids was more effective than corticosteroids alone, but a meta-analysis was not performed because of the variable response measures used in the studies. This review emphasized the need for reproducible and consistent outcome measures in future studies so that proper analyses, interpretations, and comparisons could be performed.

Orbital RT is generally well tolerated with few adverse effects, as noted in our retrospective review. With approximately 11 years of follow-up after orbital RT for GO, prior studies found no associated increase in secondary malignancy or difference in overall survival.[26,27] Although these follow-up studies were relatively small and duration of follow-up was limited, they suggest that the risk of developing a secondary malignancy due to RT is likely low. Furthermore, the risk of cataracts from low-dose RT for GO also appears low, based on prior retrospective studies that have reported similar prevalence of cataracts in patients treated with corticosteroids for GO, patients treated with low-dose RT for GO (20 Gy in 10 fractions), and the general population.[28] Again, it is recognized that it is difficult to derive the actual risk of cataracts due to RT from retrospective studies due to confounding factors that can contribute to cataract formation in the study populations (eg, age, sun exposure, corticosteroid use). An increased incidence of retinopathy has been observed in patients with existing microvascular retinopathy following radiation therapy with 20 Gy in 10 fractions.[29] Patients with both diabetes and hypertension may have a slightly increased risk of developing retinopathy, although the incidence is very small in prior studies with long-term follow-up.[26,28]

In attempts to minimize radiation-related toxicities and potentially improve treatment efficacy, alternative dose and fractionation schedules have been evaluated, including 20 Gy in 20 fractions, 10 Gy in 10 fractions, 16 Gy in 8 fractions, and 2.4 Gy in 8 fractions. Prospective studies comparing these dose and fractionation schedules have shown similar responses, but these studies are small and the endpoints for therapeutic responses were again variable.[30,31] Nonetheless, low-dose radiation has been suggested to have a greater inhibitory effect on the inflammatory response and the nitric oxide pathway, and given the greater risk of retinopathy with higher radiation doses, these studies provoke further investigation of lower dose RT for GO.

Given the retrospective nature of our study, there were limitations in the duration and completeness of clinical follow-up. The mean in-clinic follow-up was relatively short at 9.3 months, and it is possible that corticosteroid dose reductions may not have been sustained after patients were discharged. Efforts were made to capture any late increases in corticosteroid dose by providing clear instructions to return to clinic with any recurrent symptoms for patients who were discharged to their referring physician. A total of 11 patients were censored from the final data analysis due to missing documentation of corticosteroids or loss to follow-up (ie, failing to fulfill the specified criteria for planned transition of care). Smoking status, which is recognized as an important prognostic factor associated with disease activity and predictive factor for response to therapy, was not consistently available and therefore could not be included in our analyses. Finally, as baseline CAS or disease activity was not consistently reported in this cohort, the severity and natural course of disease for patients in this study may have been variable. Although the majority of our patients were referred for consideration of RT only after failing corticosteroid tapering, it is possible that a proportion of the patients may have successfully tapered off corticosteroids without RT. However, timely successful taper off corticosteroids following combined therapy without late exacerbations of orbitopathy is supported by existing literature.

Conclusions

This study suggests that orbital radiation therapy helps minimize the dose and duration of corticosteroid therapy for patients with Graves' ophthalmopathy while improving ocular symptoms. The ability to shorten the duration of corticosteroid therapy will minimize steroid-related complications and thereby improve patient quality of life. Given the clinical impact and the proposed mechanism of RT, future prospective studies should consider evaluating corticosteroid requirement as a measure of efficacy of orbital RT for GO. Consistent outcome measures and patient eligibility criteria (accounting for euthyroid state, smoking status, disease activity scored using the CAS, and incorporation of patient-reported symptoms) in future studies are crucial to meaningful interpretation of RT trials for Graves' ophthalmopathy.

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